

-- interface configuration

```
enable
configure terminal
interface <;GigabitEthernet0/0>
ip address <;ip 192.168.1.1> <mask 255.255.255.0>
no shutdown
end
copy running-config startup-config
```

-- static routing

```
enable
configure terminal
ip route <;dest NETWORK 192.168.2.0> <mask 255.255.255.0>
<next hop 10.0.0.2>
end
copy running-config startup-config
```

-- dhcp on router

```
enable
configure terminal
ip dhcp excluded-address <;range of ip s not to give out from that
network 192.168.0.1 192.168.0.10>
ip dhcp pool <;name pool>
network <;network from where to give ips 192.168.0.0> <mask
255.255.255.0>
default-router <;gate for pcs 192.168.0.1>
dns-server <;server s ip 8.8.8.8>
end
copy running-config startup-config
```

-- nat

```
interface <;the one from he lan part GigabitEthernet0/0>
ip nat inside
exit
interface <;Serial0/0/0>
ip nat outside
exit
ip access-list extended <;list name>
deny ip (to not be affected by nat) <;interval 192.168.0.0
0.0.0.127> <interval dest 192.168.0.0 0.0.0.255>
;bine ii in primu interval si vrea sa ajunga la al doilea interval nu se
aplica nat
permit ip <;interval 192.168.0.0 0.0.0.127> any
;bine ii in interval si vrea sa mearga oriunde se aplica nat
permit ip <;interval 192.168.0.0 0.0.0.255> <interval 193.231.20.0
0.0.0.255>
;bine ii in primu interval si vrea sa mearga in al doilea se aplica dat
permit ip any any ;allow everything else
ip nat inside source list <;list name> interface <outside interface
Serial0/0/0> overload
end
copy running-config startup-config
```

-- rip(dynamic routing)

```
enable
configure terminal
router rip
version 2
;advertise these networks
network <;192.168.0.0>
network <;10.0.0.0>
no auto-summary
default-information originate
end
copy running-config startup-config
```

-- stp(prevents layer-2 loops, blocks one redundant link)

```
enable
configure terminal
spanning-tree vlan 1
end
copy running-config startup-config
```

-- verification in priority order

```
show ip interface brief
show ip route
show ip nat translations
show access-lists
show spanning-tree
ping <;IP>
tracert <;IP> (PC)
```